

Shouzhuo Yang

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EDUCATION

Swarthmore College

B.A. Physics and Mathematics; GPA: 3.97/4.00

PA, USA

Sep 2019 – May 2023

Massachusetts Institute of Technology

Department of Nuclear Science and Engineering; GPA: 5.0/5.0

MA, USA

Aug 2024 – Current

COURSES AND SKILLS

Advanced Physics Coursework: String Theory, General Relativity, Quantum Field Theory, Quantum Information Science, Cosmological Physics, Condensed Matter Physics, Liquid Crystal, The Interstellar Medium

Advanced Mathematics Coursework: Algebraic Geometry, Differential Geometry, Complex Analysis, Modern Algebra, Analytical Number Theory, Stochastic Processes and Numerical Methods, Knot Theory

Languages: C/C++, Java, Python, MATLAB, The Wolfram Language

Technologies: Git, slurm, PyTorch, TensorFlow, L^AT_EX, Dedalus

PUBLICATIONS

[1] Ingle-Aynes, Kamra, L., Dai, F., J., Hou, Y., **Yang, S.**, Moodera, J. (2025). Magnetic Exchange Coupled Nonreciprocal Devices for Cryogenic Memory [*Submitted to Physics Review Letters*. arXiv:2601.06632]

[2] Ji, S., **Yang, S.**, Dominguez, W., Bester, C. Using Physics Simulations to Find Targeting Strategies in Competitive Bowling. *AIP Advances* 15, 045222 (2025)

[3] **Yang, S.**, Li, X., & Yoshida, N. (2023). Three-Dimensional Reconstruction of Weak-Lensing Mass Maps with a Sparsity Prior. II. Weighing Triaxial Cluster Halos. arXiv: 2312.00309 [Submitted to *The Astrophysical Journal*]

[4] **Yang, S.**, Zhang, B., Murdock, S. R., & Collings, P. J. (2022). Orientational order of dyes in a lyotropic chromonic liquid crystal. *Soft Matter*, 18(38), 7415-7421.

[5] **Yang, S.**, & Collings, P. J. (2020). The Genetic Algorithm: Using Biology to Compute Liquid Crystal Director Configurations. *Crystals*, 10(11), 1041.

[6] Ahmed, N., Ball, W., Buckminster, E., Rivkin, E., Torrance, D., Viscusi, J., ... & **Yang, S.** (2021). Domains of Convergence for Polyhedral Packings. arXiv preprint arXiv:2109.01289.

[7] Wang Y., Lin F., **Yang S.**, Cai P. and Igarashi S., “Efficiency Optimization of Wirelss Power Transfer System with Traction Motor Load for Modern Tram,” 2018 IEEE PELS Workshop on Emerging Technologies: Wireless Power Transfer (Wow)

RESEARCH EXPERIENCE

Superconducting Memory and Diode: Characterization and Fabrication

Research Assistant, Advisor: Dr. Jagadeesh Moodera

March 2024 – Present

- Designed and used Electron Beam Lithography to fabricate micrometer-level superconducting diode devices.
- Demonstrated single-cell write addressing using heat-assisted magnetic recording (local heating via current pulses + tiny in-plane field) while keeping adjacent cells unaffected.
- Measured strong zero-field superconducting diode effect with efficiencies $\geq 60\%$, and proposed three-state programmable diode behavior (forward / reverse / resistive) for SFQ-compatible cryogenic logic.

Modular Bootstrap for unclassified Modular Data

Massachusetts Institute of Technology

Research Assistant, Advisor: Prof. Xiao-Gang Wen

Dec 2025 – Present

- Used semi-definite programming and simpleboot to bootstrap modular data of rank 10 and 11.
- Developed hybrid DFS/SDP algorithm that iteratively tightens bounds on partition function coefficients, pruning infeasible branches to enumerate all valid integer solutions.

Generative AI for Nanomaterials Microscopy

Massachusetts Institute of Technology

Research Assistant

Aug 2024 – Jan 2026

- Benchmarked AI vs. experimental SEM across 4 magnifications using distribution-level morphology statistics (size/shape + roughness exponent from perimeter–area scaling)
- Assessed similarity via SSIM/NCC, embedding distance, pHash, IoU/Hausdorff, and ORB alignment; proposed a closed-loop digital-twin workflow for experiment planning.

Shear Estimation with Flexion

Carnegie Mellon University

Research Assistant, Advisor: Dr. Xiangchong Li and Prof. Rachel Mandelbaum

Aug 2023 – Jan 2024

- Performing image simulations, using perturbation methods to solve for flexion distortion of galaxy profiles and constraining the form of flexion’s contribution to the additive and multiplicative bias in shear estimation using spin number.
- Contributing non-affine transformation methods in public repositories BatSim and FPFs.

Machine Learning and Solar Neutrino Background in JUNO

Institute of High Energy Physics

Research Assistant, Advisor: Prof. Liangjian Wen

July 2023 – July 2024

- Using machine learning algorithms to suppress Solar Neutrino background in the $0\nu\beta\beta$ experiment in JUNO second phase.
- Implemented $0\nu\beta\beta$ and anisotropic solar neutrino generator in the JUNO offline simulation software.
- Experimenting a combination of attention mechanism and spherically invariant neural network to distinguish between solar and $0\nu\beta\beta$ signal.

3-D Cosmological Mass Map Reconstruction with Sparsity Prior

The University of Tokyo

Research Assistant, Advisor: Naoki Yoshida

Oct 2022 – May 2024

- Constructed a 3-D reconstruction algorithm to locate and weigh dark matter halo to constrain cosmological parameters.
- Implemented a parallelized lensing simulation program for different triaxial halo models (e.g., Navarro-Frenk-White Halos).
- Reached 90% detection accuracy with 5% detection mass bias for the medium mass halo with Year 1 Hyper-Suprime-Cam noise.

Genetic Algorithm to Compute Liquid Crystal Director

Swarthmore College

Research Assistant, Advisor: Peter Collings

Dec 2019 – May 2021

- Applied genetic algorithms to calculate liquid crystal director alignment by minimizing liquid crystal free energy, which includes elastic, electric, and surface free energies.
- Calculated the director alignment for Frederiks Transition, 90 Degree twisted cell, Escaped Radial Cell, and Twisted Nematic cell, before verifying solutions with Euler-Lagrange equations.
- Published the research findings in the journal *Crystals*.

Oriental Order of Dyes in Lyotropic Liquid Crystal

Swarthmore College

Research Assistant, Advisor: Peter Collings

Sep 2022 – May 2023

- Studied the physical interactions between liquid crystal molecules under mixture.
- Measured the changes in the absorption spectrum and the indices of refraction of dye molecules mixed with crystal disodium cromoglycate (DSCG).
- Showed that the alignment of the dye molecules correlates with the interaction between the dye molecules and the stacked DSCG molecules.
- Published the research findings in the journal *Soft Matter*.

Computational Plasma Research

Swarthmore College

Research Assistant, Advisor: Michael Brown

June 2020 – May 2023

- Investigated the merging of Taylor states plasma to determine whether it is suitable for inertial-magnetic confinement fusion.
- Simulated ion trajectory in the Harris Sheet, an approximation of magnetic reconnection layer, with the Boris algorithm.
- Simulated the merging of Taylor states with a resistive Magnetohydrodynamic regime under the Dedalus Framework.

AWARDS AND GRANTS

Phi Beta Kappa <i>Membership</i>	<i>2023</i>
The McWilliams Center for Cosmology Seed Grant \$4975, 250 kSU	<i>2022</i>
Swarthmore College Honors Fellowship \$5200	<i>2022</i>
Peer Assistance Certificate <i>For recognition of outstanding teaching assistant performance</i>	<i>2021</i>
The Carl Grossman Summer Opportunity Fund \$4800	<i>2021</i>
Swarthmore College Summer Research Fellowship \$4800	<i>2020</i>

TEACHING EXPERIENCE

Swarthmore College <i>Teaching Assistant for PHYS 005: Spacetime and Quanta</i>	<i>Fall 2020, 2021</i>
<i>Grader for MATH 067: Introduction to Modern Algebra</i>	<i>Spring 2020</i>
<i>Grader for MATH 035: Several-Variable Calculus with Theory</i>	<i>Spring 2020</i>
Chester Children's Chorus <i>SAT Math Tutor</i>	<i>Fall 2020</i>

LEADERSHIP AND OTHER ACTIVITIES

Swarthmore College <i>Student Hiring Committee, Department of Physics and Astronomy</i>	<i>Fall 2022</i>
<i>Student Hiring Committee, Department of Mathematics and Statistics</i>	<i>Spring 2020</i>
American Physical Society, Division of Plasma Physics <i>62nd Annual Meeting Poster Presentation</i>	<i>Nov 2020</i>
<i>63rd Annual Meeting Poster Presentation</i>	<i>Nov 2021</i>